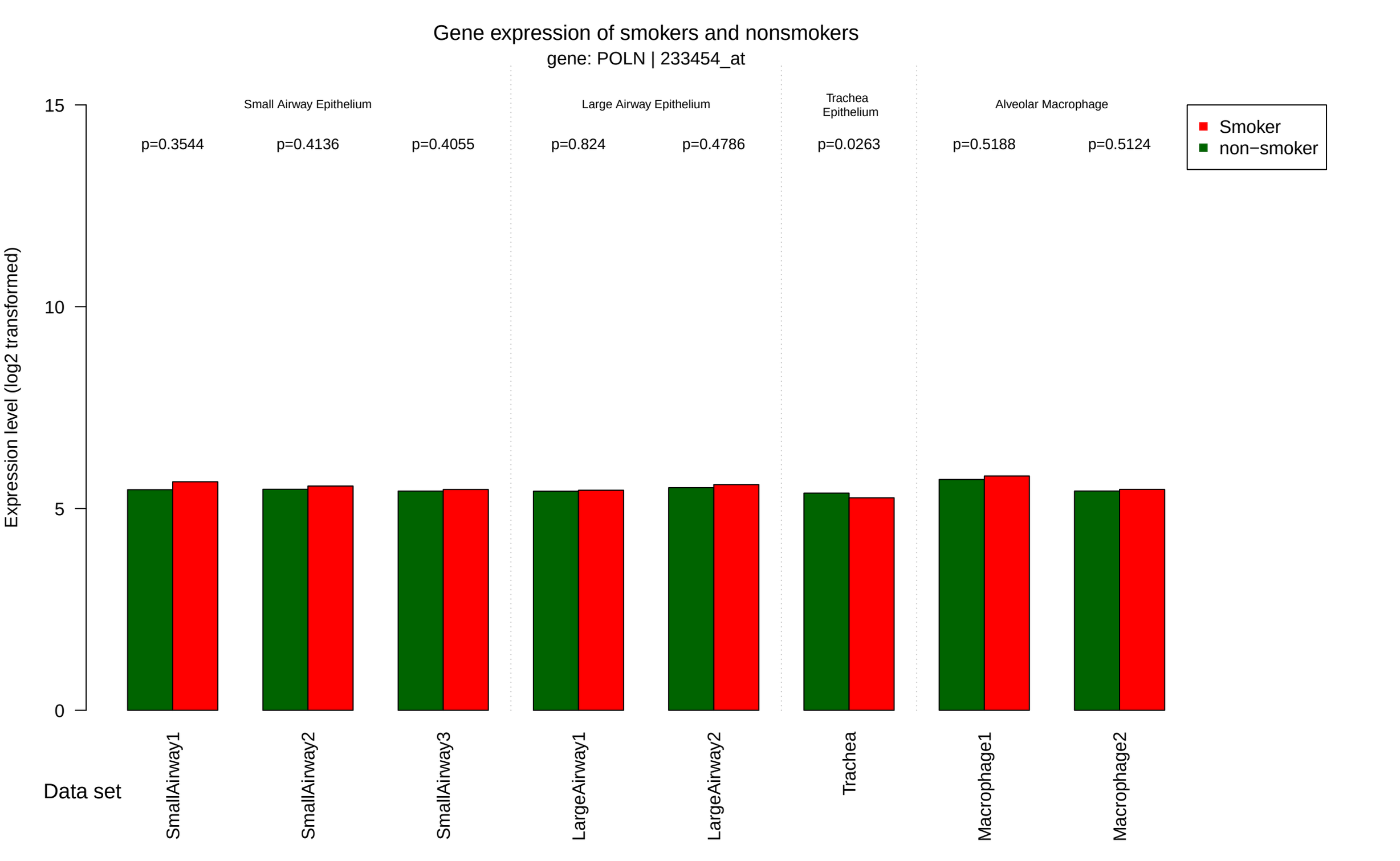
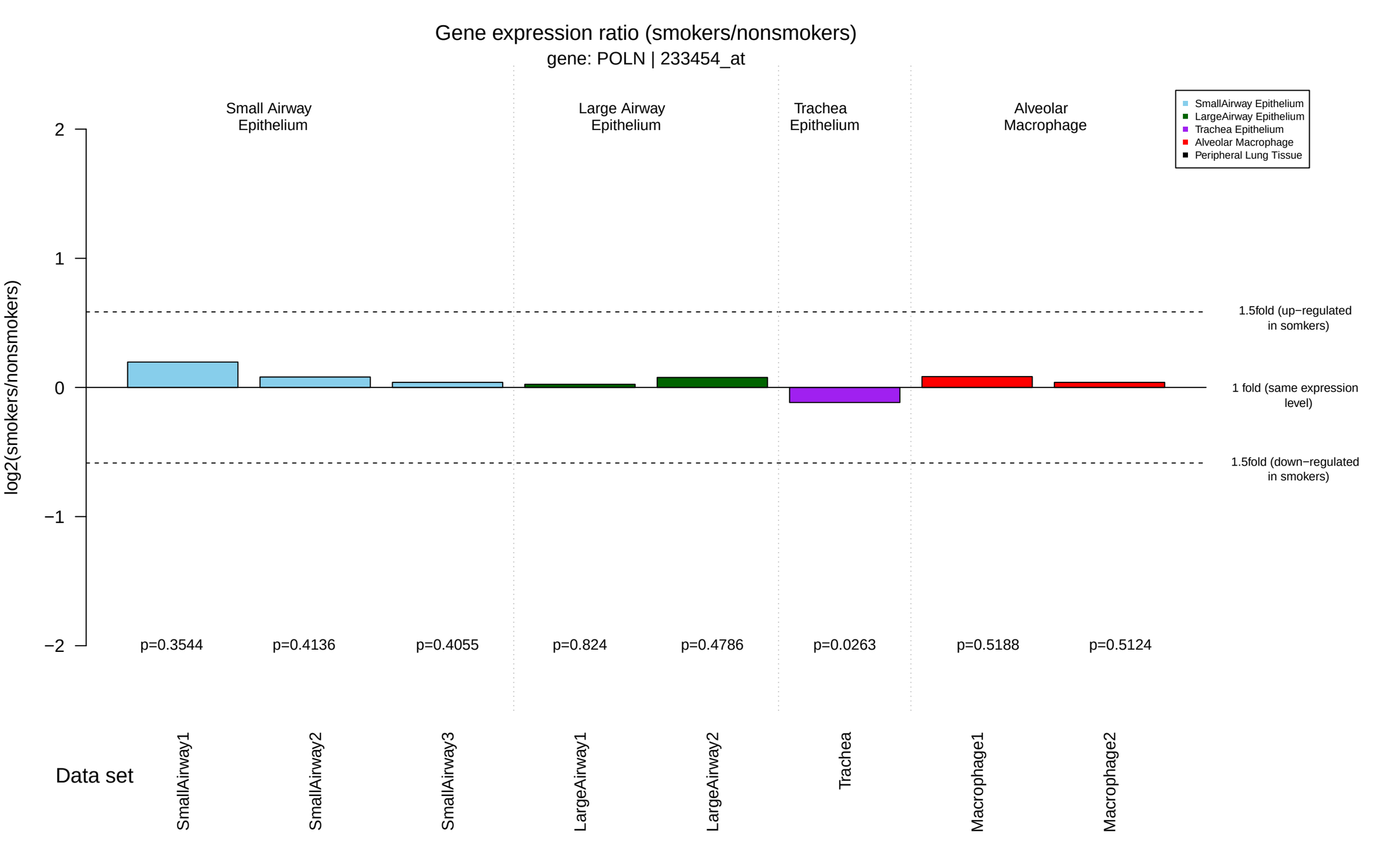
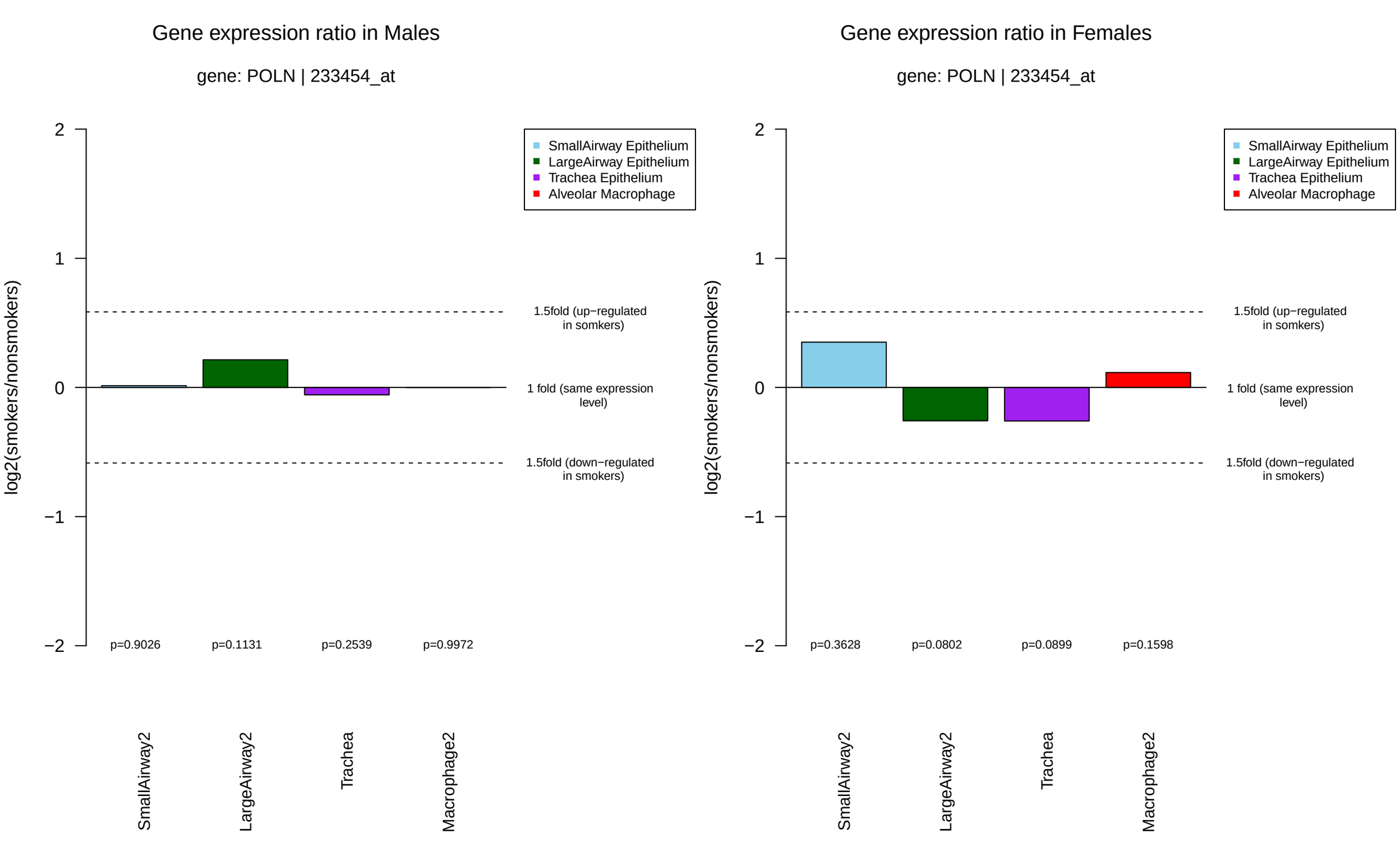
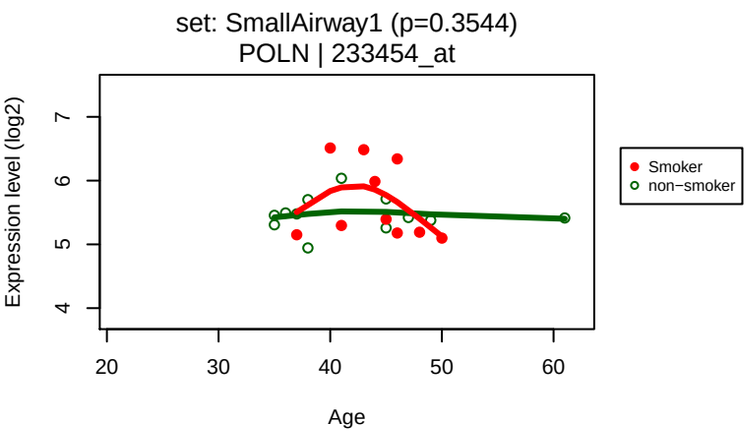
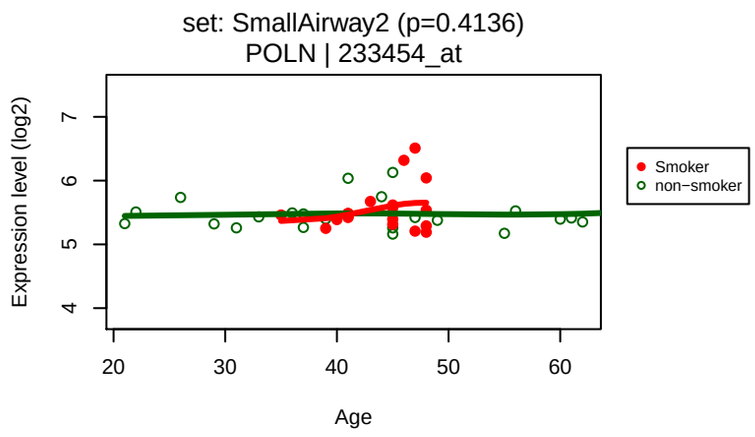
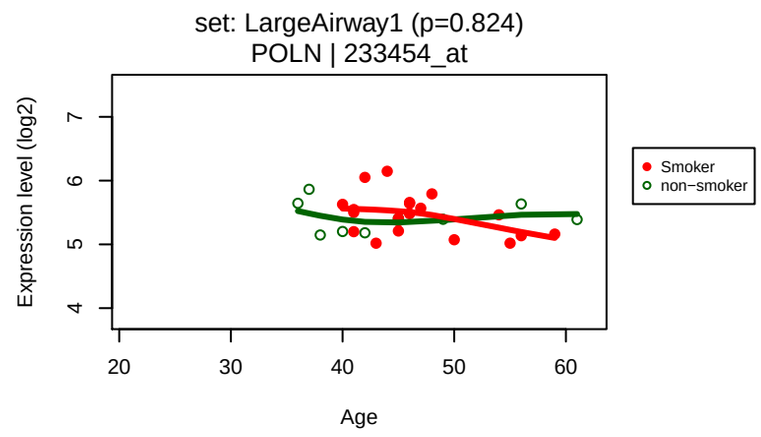

Gene expression of smokers and nonsmokers
gene: POLN | 233454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3544
p=0.4136
p=0.4055
p=0.824
p=0.4786
p=0.0263
p=0.5188
p=0.5124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

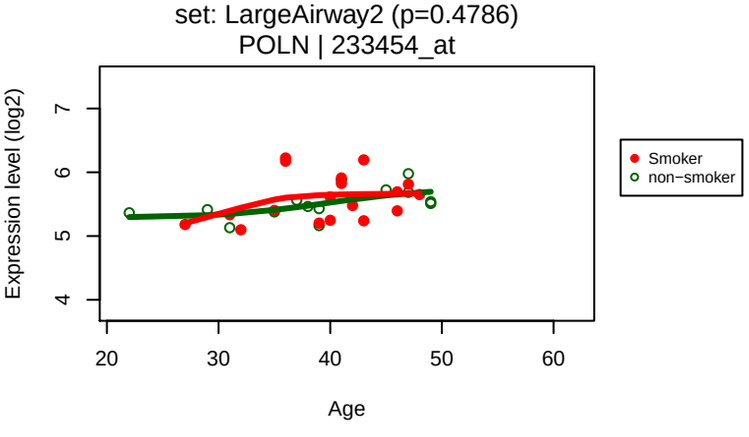

Gene expression ratio (smokers/nonsmokers)
gene: POLN | 233454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3544
p=0.4136
p=0.4055
p=0.824
p=0.4786
p=0.0263
p=0.5188
p=0.5124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

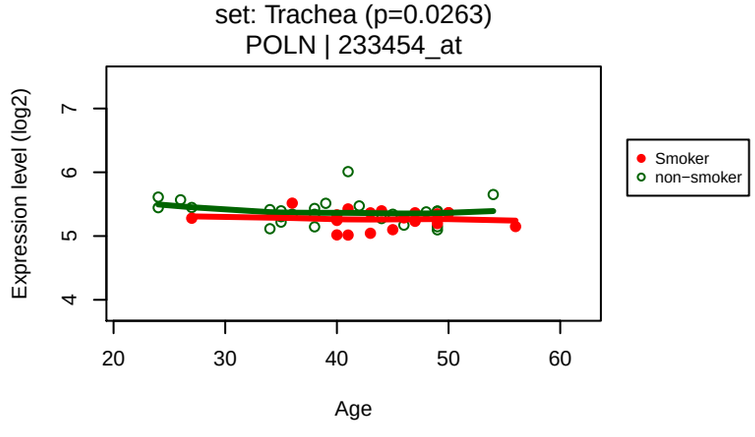

Gene expression ratio in Males
gene: POLN | 233454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9026
p=0.1131
p=0.2539
p=0.9972
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLN | 233454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3628
p=0.0802
p=0.0899
p=0.1598
SmallAirway2
LargeAirway2
Trachea
Macrophage2

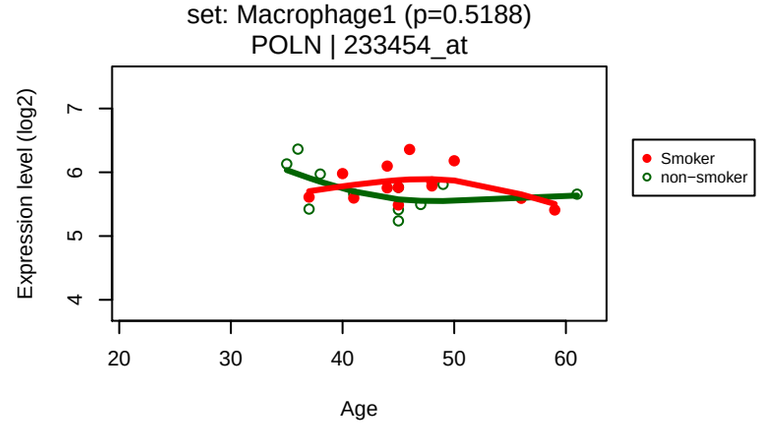

set: SmallAirway1 (p=0.3544)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker

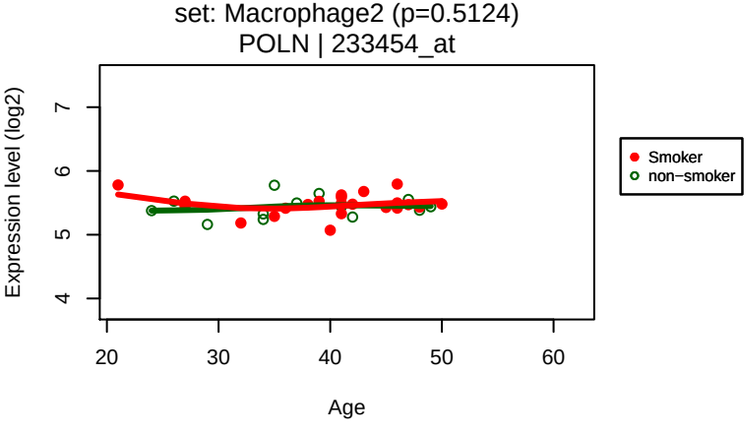

set: SmallAirway2 (p=0.4136)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker

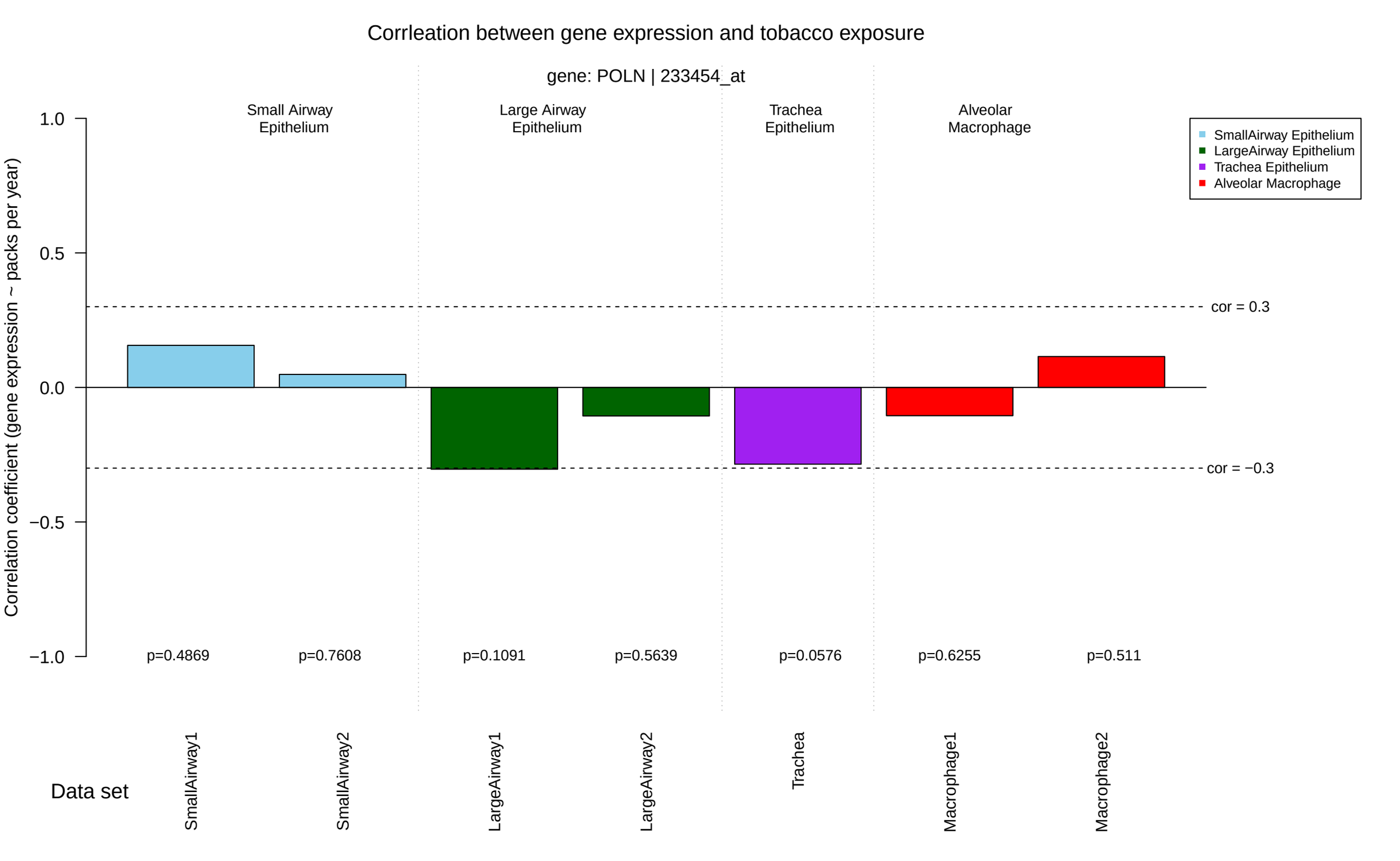

set: LargeAirway1 (p=0.824)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4786)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0263)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5188)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5124)
POLN | 233454_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLN | 233454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4869
p=0.7608
p=0.1091
p=0.5639
p=0.0576
p=0.6255
p=0.511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2